Warmup/Attendance: [Week 9: Friday 10/23/2026]

Full Name: SID:

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

For the following pairs of statements:

- Identify which is the null hypothesis and which is the alternative hypothesis.
- Write out the null hypothesis in equation form.

**1. Statement 1**

- The mean length of African elephant tusks has not changed over the last 100 years.
- The mean length of African elephant tusks has changed over the last 100 years.

**2. Statement 2**

- The rate of facial clefts is not equal for babies born to mothers who take folic acid supplements compared with those from mothers who do not.
- The rate of facial clefts is equal for babies born to mothers who take folic acid supplements compared with those from mothers who do not.

**3.** A pharmaceutical company developed a new treatment for eczema and performed a hypothesis test to see if it worked better than the company's old treatment. The P-value for the test was 10%. Which one of the following statements is true/is a valid conclusion that can be made using just this information?

A. The probability that the null hypothesis is false is 10%.

B. The P-value of about 10% was computed assuming that the null hypothesis was true.

D. The alternative hypothesis is 10 times more likely the null.